

Requirement Specification Document

E-Commerce Order Management Database System

Week 1 – Requirement Analysis and Customer Database Module

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Project Title	E-Commerce Order Management Database System
Week	Week 1

1. Introduction

1.1 Purpose

The purpose of this Software Requirement Specification (SRS) document is to define the functional and non-functional requirements of the Online Grocery Store Management System. This document serves as a guide for developers, testers, and stakeholders during software development.

1.2 Scope

The Online Grocery Store Management System allows customers to browse grocery products, register an account, place orders, make payments, and track deliveries. Administrators can manage products, categories, customers, orders, and inventory.

1.3 Objectives

- Provide an easy online shopping experience.
- Reduce manual work in grocery stores.
- Maintain customer and order records.
- Improve inventory management.
- Generate sales reports.

2. Overall Description

Product Perspective

The system is a web-based application that connects customers, administrators, and the product database.

Product Functions

- User Registration and Login
- Product Search
- Shopping Cart
- Online Payment
- Order Management
- Inventory Management
- Report Generation

User Classes

Customer

- Register and login
- Search products
- Add products to cart
- Place orders
- Make payments
- Track orders

Administrator

- Manage products
 - Manage customer information
 - Process orders
 - Update inventory
 - Generate reports
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3. Functional Requirements

FR-1 User Registration

The system shall allow customers to create an account using their personal information.

FR-2 User Login

The system shall authenticate users using username and password.

FR-3 Product Management

The administrator shall add, update, delete, and view grocery products.

FR-4 Product Search

Customers shall search products by name or category.

FR-5 Shopping Cart

Customers shall add, update, or remove products from the shopping cart.

FR-6 Order Management

The system shall allow customers to place orders and administrators to process them.

FR-7 Payment

The system shall support secure online payment methods.

FR-8 Delivery Tracking

Customers shall be able to view the status of their orders.

4. Non-Functional Requirements

Performance

- The system should load within 3 seconds.
- Support multiple users simultaneously.

Security

- Passwords must be encrypted.
- Secure login authentication.
- Protect customer information.

Reliability

- The system should be available 24×7.
- Data should be backed up regularly.

Usability

- User-friendly interface.
- Easy navigation.

Maintainability

- Easy to update and maintain.
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5. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or above
- Hard Disk: 100 GB
- Internet Connection

Software Requirements

- Operating System: Windows 10/11
 - Database: MySQL
 - Front-end: HTML, CSS, JavaScript
 - Back-end: Java/PHP
 - Browser: Google Chrome or Microsoft Edge
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6. Database Modules

Customer Table

- Customer ID
- Name
- Email
- Phone Number
- Address
- Password

Product Table

- Product ID
- Product Name
- Category
- Price
- Quantity

Order Table

- Order ID
 - Customer ID
 - Product ID
 - Order Date
 - Total Amount
 - Order Status
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7. Assumptions

- Users have internet access.
 - The payment gateway is available.
 - Customers provide correct information during registration.
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8. Constraints

- Stable internet connection is required.
 - Only registered users can place orders.
 - Admin access is restricted.
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9. Future Enhancements

- Mobile Application
 - AI-based Product Recommendation
 - Voice Search
 - Multiple Language Support
 - Digital Wallet Integration
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10. Conclusion

The Online Grocery Store Management System provides an efficient and user-friendly platform for purchasing groceries online. It simplifies shopping for customers while helping administrators manage products, inventory, and orders effectively. This SRS document

serves as the foundation for designing, developing, testing, and maintaining the software successfully.